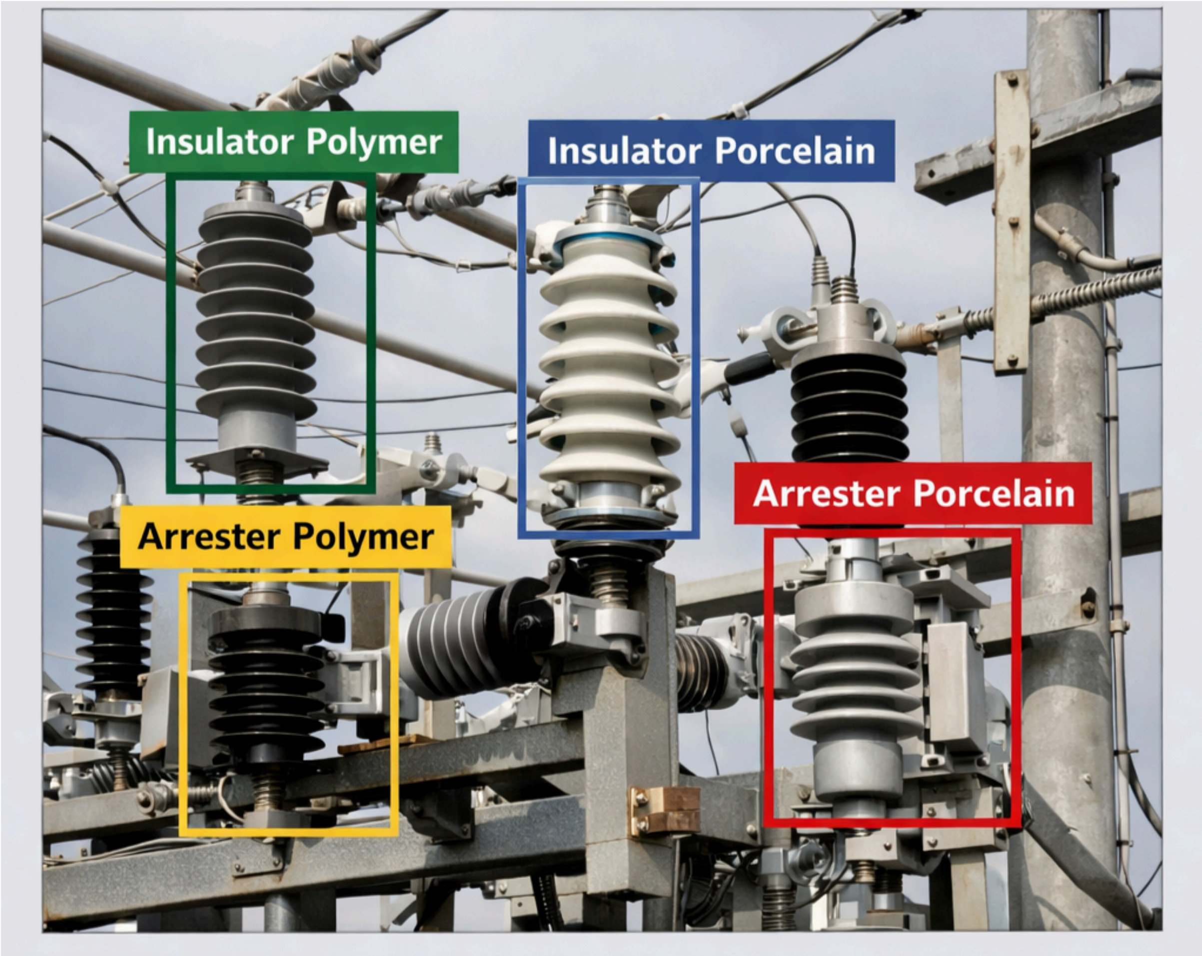
Electrical Engineering Test Guidelines

You need to do the following steps:

1. Go to https://app.cvat.ai/ and log in. If you don't have an account, create one.
2. After authorisation, create a new task with the following parameters:

    - Name: Electrical Task
    - Create and use **only the following four labels** exactly as written:
        1. **Insulator Polymer**
        2. **Insulator Porcelain**
        3. **Arrester Polymer**
        4. **Arrester Porcelain**

⚠️ Do not create additional labels or modify label names.



Add images from the **"Images for annotation"** folder, which is in the same archive

3. Open the created task and annotate all images according to the rules described in the Task Requirement section below

4. Save the task and provide a link to the task in this form

https://forms.gle/ooHL8WkX6s3xg3Y46

Documentation:

Account registration:

https://opencv.github.io/cvat/docs/manual/basics/registration/
Task creation:
https://opencv.github.io/cvat/docs/manual/basics/create_an_annotation_task/
Shape Tool :
https://docs.cvat.ai/docs/annotation/manual-annotation/shapes/shape-mode-basics/
How to draw a bounding box:
https://www.youtube.com/watch?v=TDae8RkGbKk7IBG&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:c20f2639,vid:TDae8RkGbKk,st:0
CVAT.ai Product Tour #20: Improve Annotation Speed and Accuracy with Layers

NB: IF YOU ARE NEW TO CVAT, PLEASE WATCH THE YOUTUBE VIDEO BELOW TO UNDERSTAND HOW TO USE

CVAT - Computer Vision Annotation Tool - Open Data Annotation Platform for Image & …

Task Instructions

1. **Identify all visible insulators and surge arresters** in the provided image.
2. **Draw a tight bounding box** around each individual component.
    - The bounding box should fully cover the component.
    - Avoid including surrounding structures (poles, beams, wires) unless they are inseparable from the object.
3. **Classify each component** based on:
    - **Function:** Insulator vs Surge (Lightning) Arrester
    - **Material:** Polymer vs Porcelain
4. **Assign the correct label** from the class list to each bounding box.

Annotation Rules

- Annotate **every eligible object** in the image.
- Do **not annotate partial, broken, or unclear objects**.
- Do **not overlap bounding boxes unnecessarily**.

- Each object must have **only one bounding box and one label**.
- Ensure labels are **accurate and consistent** across the image.

Identification Guidelines (Domain Knowledge Check)

Use the following characteristics to guide your classification:

1. **Insulator Polymer**
   Polymer (composite / silicone rubber) insulators

   Dark grey or black housing

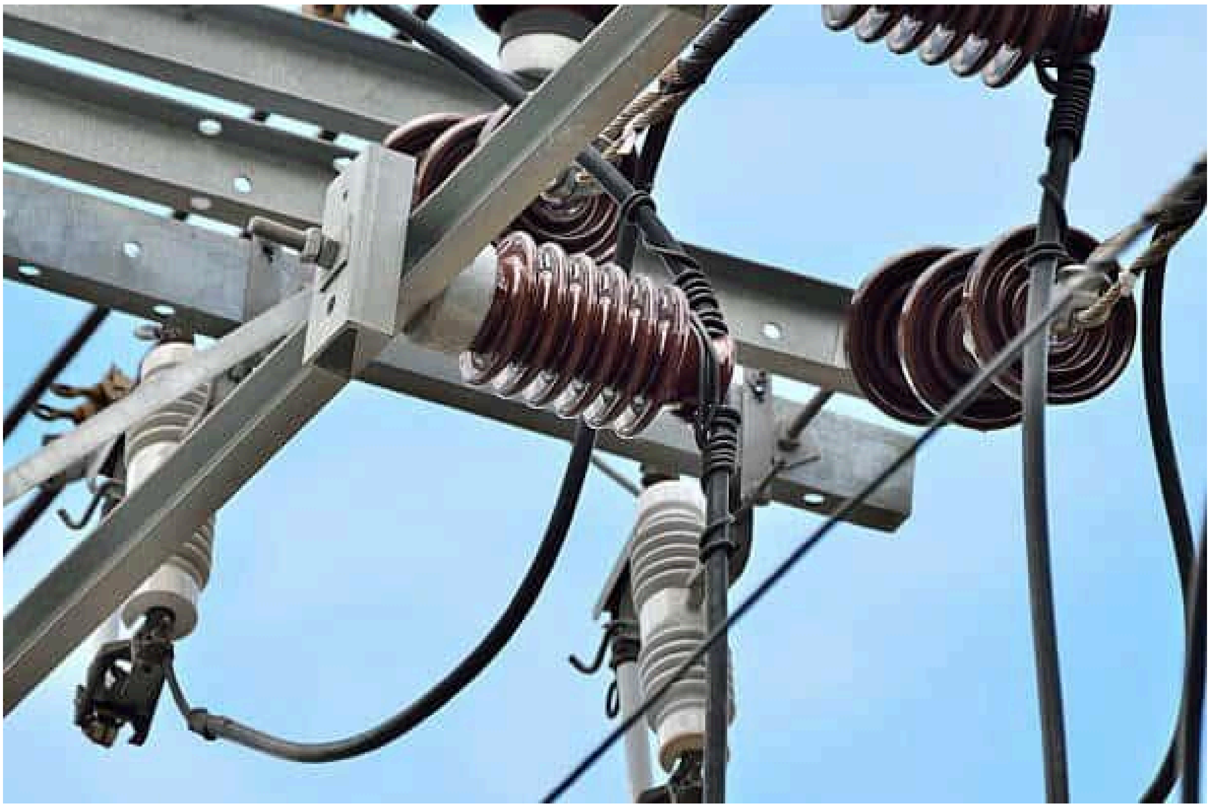
   Flexible, non-ceramic sheds

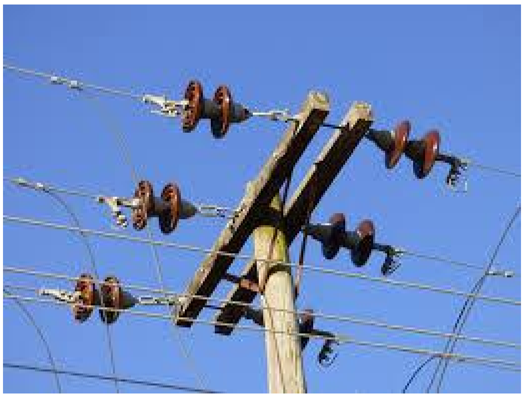

2. **Insulator Porcelain**

    Porcelain / ceramic insulators

    Light grey or off-white color

    Rigid, glossy ceramic sheds






(a)
(c)
(b)
(d)
(e)

3. Arrester Polymer

Polymer-housed surge (lightning) arresters

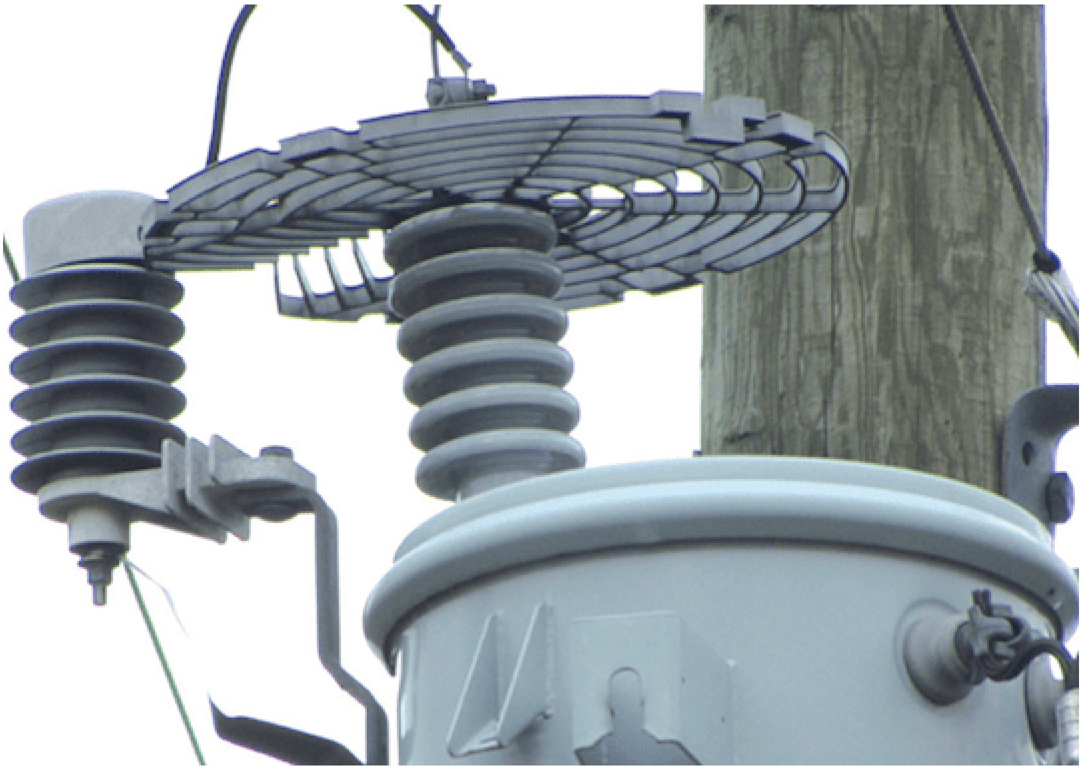
Dark grey or black rubber housing
Typically connected between phase and ground

4. Arrester Porcelain

Porcelain-housed surge (lightning) arresters

Light grey ceramic housing

Heavier, segmented ceramic sheds

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